

2016.0 RANGE ROVER (LG), 303-06

# STARTING SYSTEM – TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

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## PRINCIPLES OF OPERATION

For a detailed description of the starting system and operation, refer to the relevant Description and Operation section of the workshop manual. REFER to: [Starting System](#) (303-06A Starting System - TDV6 3.0L Diesel, Description and Operation).

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## INSPECTION AND VERIFICATION

### CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

### NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.

2. Visually inspect for obvious signs of mechanical and electrical damage.

### Visual Inspection

| MECHANICAL                                                                                      | ELECTRICAL                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>▪ Starter motor</li><li>▪ Engine (turns freely)</li></ul> | <ul style="list-style-type: none"><li>▪ Battery</li><li>▪ Fuses</li><li>▪ Starter relay</li><li>▪ Wiring harness(es)</li><li>▪ Damaged, loose or corroded connectors</li><li>▪ Ignition switch</li><li>▪ Generator</li><li>▪ Transmission control module</li><li>▪ Engine control module</li></ul> |

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSM's which may be valid for the specific customer complaint and carry out the recommendations as needed.

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## SYMPTOM CHART

| SYMPTOM                             | POSSIBLE CAUSES                                                                                                                | ACTION                                                                                                                                                                                                         |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The engine does not crank (starter) | <ul style="list-style-type: none"><li>▪ Gear selector not in <b>P</b> or <b>N</b> position (vehicles with automatic)</li></ul> | Make sure the gear selector is in the <b>P</b> or <b>N</b> position and correctly adjusted. Check the battery condition and state of charge. Check for DTCs indicating an immobilizer fault. Check the starter |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| motor does <b>not</b> turn)                                | transmission)<br><ul style="list-style-type: none"> <li>▪ Battery</li> <li>▪ Starter relay</li> <li>▪ Invalid key code received by Body Control Module (BCM)</li> <li>▪ Harness/connectors</li> <li>▪ Starter motor</li> <li>▪ Ignition switch</li> <li>▪ Generator</li> <li>▪ Transmission control module</li> <li>▪ Engine control module</li> <li>▪ Engine seized</li> </ul> | motor relay, ignition switch and generator circuits. Refer to the electrical guides. Check for TCM and ECM DTCs. Check that the engine turns freely                                                            |
| The engine does not crank (starter motor <b>does</b> turn) | <ul style="list-style-type: none"> <li>▪ Starter motor installation</li> <li>▪ Starter motor</li> <li>▪ Flywheel/drive plate ring gear</li> </ul>                                                                                                                                                                                                                               | Check the starter motor installation (fasteners tight, starter motor square to engine, etc). Check the flywheel/drive plate ring gear teeth for damage, foreign objects, etc                                   |
| Engine cranks too slowly                                   | <ul style="list-style-type: none"> <li>▪ Battery</li> <li>▪ Harness/connectors</li> <li>▪ Starter motor</li> <li>▪ Oil grade</li> </ul>                                                                                                                                                                                                                                         | Check the battery condition and state of charge. Check the starter motor circuits. Refer to the electrical guides. Check the engine oil grade and condition                                                    |
| Engine cranks too fast                                     | <ul style="list-style-type: none"> <li>▪ Low engine compression</li> </ul>                                                                                                                                                                                                                                                                                                      | Check the engine compressions                                                                                                                                                                                  |
| Excessive starter motor noise                              | <ul style="list-style-type: none"> <li>▪ Starter motor</li> <li>▪ Flywheel/drive plate ring gear</li> <li>▪ Starter motor installation/casing</li> </ul>                                                                                                                                                                                                                        | Check the starter motor installation (fasteners tight, motor square to engine, etc). Check the starter motor casing condition. Check the flywheel/drive plate ring gear teeth for damage, foreign objects, etc |
| ECO (stop/start)                                           | <ul style="list-style-type: none"> <li>▪ ECO system inhibited</li> </ul>                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                |

|                    |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| system inoperative | <ul style="list-style-type: none"> <li>ECO system fault</li> </ul> | <p><b>NOTE:</b></p> <div> <p>"ECO STOP/START FAULT" will be displayed in the instrument cluster message centre when the system is inoperative due to a fault.</p> </div> <ul style="list-style-type: none"> <li>Refer to the ECO System Inhibits table for possible inhibit conditions           <ul style="list-style-type: none"> <li>ECO system inhibited due to owner error: Explain/demonstrate the ECO system operation to the owner</li> <li>ECO system inhibited due to battery condition: Using the manufacturer approved diagnostic system, perform the Power Supply Service Mode Diagnostic routine</li> </ul> </li> <li>Using the manufacturer approved diagnostic system, check the Engine Control Module (ECM) for related DTCs and refer to the relevant DTC index</li> </ul> |
|--------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ECO SYSTEM INHIBITS

The inhibit conditions listed below will not set Diagnostic Trouble Codes (DTCs) but will set flags which can be viewed using the manufacturer approved diagnostic system - Datalogger / Engine Control Module (ECM). Normal ECO system operation will resume when an inhibit condition is no longer present, but the battery related inhibits may persist for some time.

### NOTE:

Do not reset the Battery Monitoring System (BMS) to rectify battery condition inhibits. This may temporarily restore ECO system operation but will lead to a recurrence of the fault.

## INHIBITS

- ECO switch has been pressed by driver
- Transmission not in neutral
- Clutch pedal not fully released
- Accelerator pedal not released
- Hood open detected
- Driver door detected open
- Driver seatbelt not fastened
- Hill Descent Control (HDC) is switched on
- Terrain response special programme selected
- Climate control demand exceeds calibrated threshold
- Heated windshield operating
- Trailer connected
- Brake booster vacuum below threshold
- Engine coolant temperature below threshold
- Engine oil temperature below threshold
- Catalytic converter outside either pre or post calibration range
- Diesel Particulate Filter (DPF) active regeneration in progress (diesel engines only)
- Battery state of charge low
- Battery cold cranking capability below threshold
- Ambient temperature above 40°C
- Ambient temperature below 0°C

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## D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5](#),  
[DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).

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**NOTE:**

This test will require the technician to listen for audible result, high surrounding noise may hinder the outcome

**WARNING:**

During this test be aware of hot or moving parts. Failure to follow this instruction may result in personal injury

**NOTES:**

- You will require assistance during this test
- You may require extended test leads for the multimeter

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During this test be aware of hot or moving parts. Failure to follow this instruction may result in personal injury

**NOTE:**

You will require assistance during this test

**WARNING:**

During this test isolate the vehicle battery. Failure to follow this instruction may result in personal injury

**PINPOINT TEST A : STARTER MOTOR PRIOR APPROVAL TEST PROCEDURE****TEST  
CONDITIONS****DETAILS/RESULTS/ACTIONS****A1: PRE CHECKS (MANUFACTURER APPROVED BATTERY TESTER)**

- 1 Please refer to workshop manual section 414-01 and carry out the diagnosis and testing using the manufacturer approved test

|  |                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|  | equipment. Record the information using the battery care report sheet                                                                                                                                                                                                                                                                                                                                                    |
|  | <p>Have the battery care routines using the manufacturer approved test equipment been carried out and has the information using the battery care report sheet been recorded?</p> <p><b>Yes</b><br/>Proceed to next step</p> <p><b>No</b><br/>Carry out the battery care routines using the manufacturer approved test equipment. Record the information using the battery care report sheet<br/>Proceed to next step</p> |

## A2: PRE CHECKS (GENERAL WEATHER CONDITIONS)

|  |                                                                                                                                                                                                                                                                                                                                          |
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|  | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <p>The results of this test step may be required by dealer technical support</p> </div> <p><b>1</b> Confirm with the customer and record the weather conditions and ambient temperature (as accurately as possible) at the time of failure</p> |
|  | <b>2</b> Confirm and record the weather conditions and ambient temperature (as accurately as possible) at the time of test                                                                                                                                                                                                               |
|  | <p>Have the weather conditions and ambient temperature been recorded</p> <p><b>Yes</b><br/>Proceed to next step</p> <p><b>No</b><br/>Record the weather conditions and ambient temperature<br/>Proceed to next step</p>                                                                                                                  |

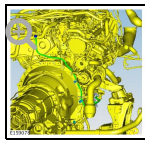
## A3: CHECK FOR RELATED STARTER MOTOR, IMMOBILIZER DIAGNOSTIC TROUBLE CODES

|  |                                                                                                                                                                                               |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, check for starter motor or immobilizer, related DTCs and refer to the relevant DTC index                                          |
|  | <p>Are starter motor or immobilizer, related DTCs logged?</p> <p><b>Yes</b><br/>Carry out any repair action as requiredClear the DTC and retest</p> <p><b>No</b><br/>Proceed to next step</p> |

## A4: CHECK INSTALLATION OF STARTER MOTOR BREATHER TUBE

|  |                                                                                                                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> If a vent-cap breather tube is installed, check this is routed correctly and that the cap is secure. If the cap is insecure or missing, or if there is damage to the breather tube, then replace the missing/damaged components as required. If a shepherd's-crook type breather tube is |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|





installed, replace the breather tube with a vent-cap breather tube

Is the starter motor breather tube installed and routed correctly?

**Yes**

Proceed to next step

**No**

Install the breather tube and route it correctly. Attempt 10 starts to the vehicle. If the vehicle does **NOT** start during any of the 10 attempts. Install new starter motor. If the vehicle **DOES** start during all of the 10 attempts. Proceed to next step

#### A5: STARTER MOTOR CHECKS

1 Set the ignition ON

2 Depress the brake pedal

3 Depress the start button

Was there an audible click when the start button was pressed and did the vehicle start?

(Audible click will not be heard if the vehicle starts)

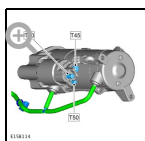
**Yes**

Proceed to next step

**No**

Set the ignition OFF. Repeat this starter motor check a further 10 times. Record the result for number of none click tests. If there was an audible click when the start button was pressed during the 10 test cycle. Proceed to next step

#### A6: CHECK CONNECTIONS ON THE STARTER MOTOR



1 Please refer to workshop manual section 303-06 specification section

2 Gain access to the starter motor

3 Refer to the electrical circuit diagrams and check the battery power supply and terminal to the starter motor (battery positive cable nut T30 permanent feed B1)

4 Refer to the electrical circuit diagrams and check the operating supply and terminal to the starter motor (switch lead to the starter motor solenoid nut T50)

5 Refer to the electrical circuit diagrams and check the power feed stud

|  |                                                                                                                                                                                                                                                          |
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|  | on the starter motor (T45 terminal)                                                                                                                                                                                                                      |
|  | <b>6</b> Refer to the electrical circuit diagrams and inspect connections for signs of becoming loose, water ingress, and terminals/stud for damage and/or corrosion                                                                                     |
|  | <p>Are terminal/stud connections to starter motor, loose, damaged or corroded?</p> <p><b>Yes</b><br/>Rectify as required<br/>Re-check the system is operating correctly<br/>Return vehicle to the customer</p> <p><b>No</b><br/>Proceed to next step</p> |

#### A7: VOLTAGE DROP GROUND CIRCUIT

|  |                                                                                                                                                                                                                                                                                                                                                                                  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Connect the multimeter between the battery negative terminal and the starter motor ground (where bolted to the bell housing)                                                                                                                                                                                                                                            |
|  | <b>2</b> Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading                                                                                                                                                                                                                                    |
|  | <p>Is reading equal to or below 0.6 volts?</p> <p><b>Yes</b><br/>Proceed to next step</p> <p><b>No</b><br/>Refer to the electrical circuit diagrams and check ground circuit between battery negative terminal and starter motor for corrosion and high resistance<br/>Rectify as required<br/>Re-check the system is operating correctly<br/>Return vehicle to the customer</p> |

#### A8: VOLTAGE DROP POWER CIRCUIT

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Connect the multimeter between the battery positive terminal and the power feed stud on the starter motor (T45 terminal)                                                                                                                                                                                                                                                                                                               |
|  | <b>2</b> Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading                                                                                                                                                                                                                                                                                                   |
|  | <p>Is reading equal to or below 0.8 volts?</p> <p><b>Yes</b><br/>Proceed to next step</p> <p><b>No</b><br/>Refer to the electrical circuit diagrams and check power circuit between battery positive terminal and the power feed stud on the starter motor (T45 terminal) including mega fuses, for corrosion and high resistance<br/>Rectify as required<br/>Re-check the system is operating correctly<br/>Return vehicle to the customer</p> |

#### A9: VOLTAGE AT STARTER MOTOR (T50)

|  |                                                                                     |
|--|-------------------------------------------------------------------------------------|
|  | <b>1</b> Check the operating supply voltage to the starter motor (T50)              |
|  | <b>2</b> Connect the multimeter between the starter motor (T50) terminal and ground |

|  |                                                                                                                                                                                                                                                                                                                                       |
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|  | <b>3</b> Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading                                                                                                                                                                                         |
|  | <p>Is the reading between 9 &amp; 12 volts during cranking?</p> <p><b>Yes</b><br/>Proceed to next step</p> <p><b>No</b><br/>Refer to the electrical circuit diagrams and check starter relay circuit for open circuit, high resistanceRectify as requiredRe-check the system is operating correctlyReturn vehicle to the customer</p> |

#### A10: VOLTAGE AT STARTER MOTOR SOLENOID (T45)

|  |                                                                                                                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Check the operating supply voltage to the starter motor (T45)                                                                                             |
|  | <b>2</b> Connect the multimeter between the starter motor (T45) terminal and ground                                                                                |
|  | <b>3</b> Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading                      |
|  | <p>Is the reading between 6 &amp; 9 volts during cranking?</p> <p><b>Yes</b><br/>Proceed to next step</p> <p><b>No</b><br/>Check and install new starter motor</p> |

#### A11: STARTER MOTOR VISUAL INSPECTION

|  |                                                                                                                                                                                                                                                                                                                                  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Remove the starter motor                                                                                                                                                                                                                                                                                                |
|  | <b>2</b> Visually inspect the starter motor ring gear for failure                                                                                                                                                                                                                                                                |
|  | <b>3</b> Visually inspect the flywheel ring gear for failure                                                                                                                                                                                                                                                                     |
|  | <p>Was there evidence of failure to the starter motor of flywheel ring gear</p> <p><b>Yes</b><br/>Check and install new parts as requiredRe-check the system is operating correctlyReturn vehicle to the customer</p> <p><b>No</b><br/>Contact dealer technical support with all of the recorded values from the above tests</p> |

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).

